



Humanitarian Air Cargo Optimization Using BIP

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Project Overview

This project focuses on solving a real-world logistics optimization problem using Binary Integer Programming (BIP).

A humanitarian organization needs to transport emergency supplies using a cargo aircraft after a disaster such as:

- Earthquakes
- Floods
- Wars
- Hurricanes

The aircraft has limited:

- Carrying weight
- Cargo space
- Transportation budget
- Refrigeration slots

The organization must decide which items should be loaded into the aircraft to maximize the overall humanitarian benefit while satisfying all operational constraints.

The project uses:

- Mathematical Modeling
 - Operations Research
 - Optimization Techniques
 - Python Programming
 - Integer Linear Programming
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Real-World Motivation

In real humanitarian operations:

- Resources are limited
- Aircraft capacity is limited
- Emergency supplies are critical
- Transportation cost matters
- Medical supplies may require cooling
- Fragile supplies need careful handling

Choosing the wrong shipment combination can:

- Waste resources
- Reduce rescue efficiency
- Increase costs
- Delay aid delivery

Optimization helps decision-makers select the best shipment plan mathematically instead of manually guessing.

Problem Type

This is an **Advanced Multi-Constraint Knapsack Problem**.

The classical knapsack problem only considers value and weight. This project extends it into a more realistic logistics optimization model by adding:

- Volume constraints
- Budget constraints
- Category requirements
- Refrigeration limits
- Fragile item limits
- Minimum shipment requirements

This makes the project significantly more advanced and practical.

Objective of the Project

The main objective is to **Maximize Humanitarian Utility** while satisfying all operational constraints.

The utility represents:

- Item importance
- Emergency priority
- Operational usefulness

The optimization model determines which items to include and which to exclude to achieve the highest possible utility.

Dataset Explanation

The project contains **8 available cargo items**. Each item has several characteristics:

Feature	Meaning
Utility	Humanitarian importance
Weight	Aircraft weight usage
Volume	Cargo space usage
Cost	Transportation/handling cost
Priority	Emergency importance
Fragile	Damage sensitivity
Cold Storage	Needs refrigeration
Category	Medical, Food, or Tools

Example Item — Item B

Attribute	Value
Utility	80
Weight	20
Volume	3
Cost	40
Fragile	Yes
Cold Storage	Yes
Category	Medical

Meaning: highly valuable medical supply that requires refrigeration and consumes significant aircraft resources.

Decision Variables

The model uses binary decision variables. For each item:

$$x_i = \begin{cases} 1 & \text{if item is selected} \\ 0 & \text{otherwise} \end{cases}$$

Example: - $x_A = 1 \rightarrow$ Item A is loaded - $x_A = 0 \rightarrow$ Item A is not loaded

Because decisions are YES/NO decisions, binary variables are ideal.

Objective Function

The objective function maximizes total humanitarian utility:

$$\max Z = 60x_A + 80x_B + 90x_C + 75x_D + 30x_E + 75x_F + 60x_G + 30x_H$$

This equation calculates the total benefit of selected items. The solver tries to maximize this value.

Constraints

Constraints represent real operational limitations.

1. Weight Constraint

Aircraft cannot exceed maximum carrying weight:

$$10x_A + 20x_B + 30x_C + 15x_D + 5x_E + 18x_F + 12x_G + 8x_H \leq 65$$

2. Volume Constraint

Cargo space is limited:

$$2x_A + 3x_B + 4x_C + 3x_D + x_E + 3x_F + 2x_G + 2x_H \leq 12$$

3. Budget Constraint

Transportation and handling costs are limited:

$$25x_A + 40x_B + 55x_C + 20x_D + 10x_E + 35x_F + 30x_G + 15x_H \leq 120$$

4. Medical Requirement Constraint

At least two medical items are required:

$$x_A + x_B + x_C \geq 2$$

5. Food Requirement Constraint

At least one food package must be included:

$$x_D + x_E \geq 1$$

6. Fragile Item Constraint

Too many fragile items increase damage risk (maximum two allowed):

$$x_A + x_B + x_F \leq 2$$

7. Refrigeration Constraint

Only limited refrigerated cargo slots are available:

$$x_B + x_C \leq 2$$

8. Minimum Shipment Size

At least four items must be included:

$$x_A + x_B + x_C + x_D + x_E + x_F + x_G + x_H \geq 4$$

Why Binary Integer Programming?

Binary Integer Programming is suitable because:

- Decisions are discrete
- Items are either selected or not selected
- Fractional selection is impossible

For example, selecting 0.5 of a medical package makes no sense. Thus, variables must be binary.

Why PuLP?

The project uses the **PuLP** library in Python because it:

- Is simple to use
 - Supports integer programming
 - Interfaces with the CBC solver
 - Is widely used in optimization education
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Solver Used

The project uses the **CBC MILP Solver** (Mixed Integer Linear Programming).

The solver:

- Reads constraints
 - Evaluates possible solutions
 - Searches for the optimal combination
 - Guarantees the best solution
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Output Explanation

The solver returns:

- Optimal status
- Selected items

- Maximum utility
- Total weight used
- Total volume used
- Total budget used

Example Optimal Solution

Selected Items: A, B, D, E, H

Project Complexity

This project demonstrates several advanced optimization concepts:

Concept	Included
Binary Variables	✓
Linear Objective Function	✓
Multiple Constraints	✓
Integer Programming	✓
Resource Allocation	✓
Logistics Optimization	✓
Real-World Modeling	✓

Applications of This Model

This type of optimization is used in:

- Airlines
- Shipping companies
- Military logistics
- Disaster relief organizations
- Supply chain companies
- Warehouse management
- Humanitarian operations

Examples: UN humanitarian aid, Red Crescent logistics, military transport planning, Amazon cargo optimization.

Future Improvements

Multi-Aircraft Optimization

Extend the model to support multiple planes or trucks.

Route Optimization

Combine cargo selection with delivery routes.

Multi-Objective Optimization

Maximize utility while simultaneously minimizing cost and emissions.

AI Algorithms

Compare exact optimization with heuristic approaches: - Genetic Algorithms - Simulated Annealing - Reinforcement Learning

Real-Time Dynamic Optimization

Adapt decisions dynamically during active emergencies.

Technologies Used

Technology	Purpose
Python	Programming
PuLP	Optimization modeling
CBC Solver	Integer optimization
Integer Programming	Mathematical optimization

Conclusion

This project successfully models a real-world humanitarian logistics problem using Binary Integer Programming. The optimization model maximizes humanitarian utility, respects all operational constraints, and selects the best shipment combination mathematically. It demonstrates how Operations Research and Optimization Techniques can improve real logistics decision-making in critical humanitarian scenarios.